From Idea to Implementation

Christopher Bolger
3rd Year Computer Science
Game Development

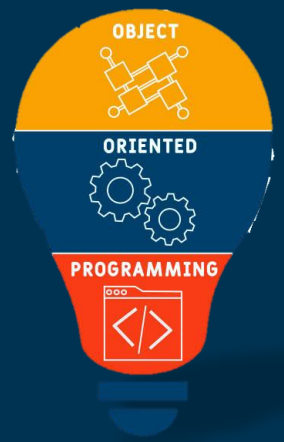
Resources

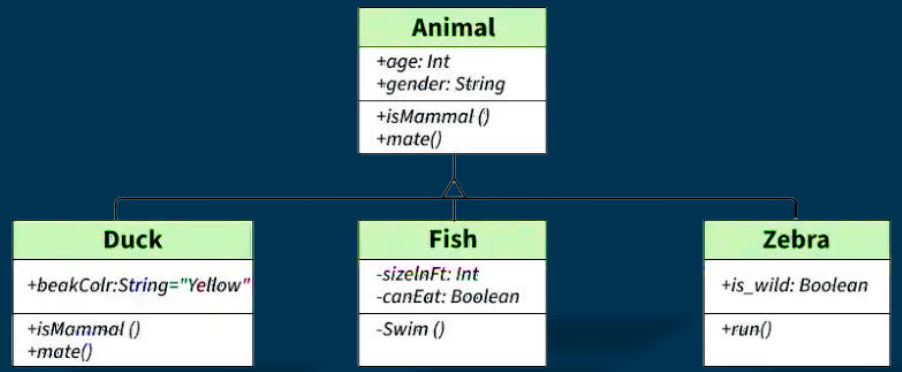
Slides available to download:

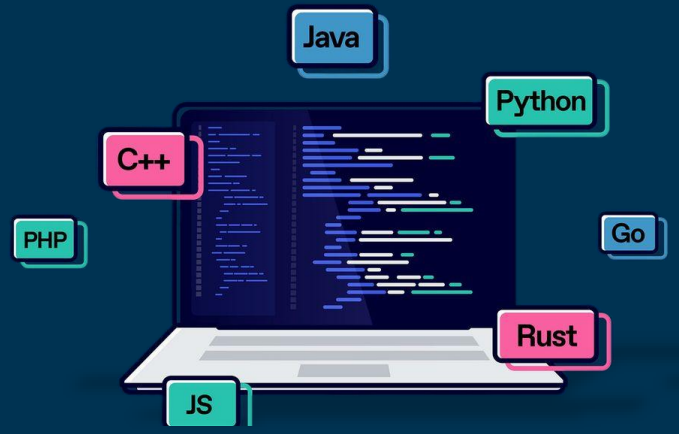
https://github.com/christopher-bolger/UMLWorkshop



Workshop Aim







Identify requirements from a brief

Design objects

Create relationships between them

What is OOP?

Object-Orientated Programming

Object-Orientated Structure



Class

Blueprint for an item



Object

The specific item



Method

An item's function(s)



Attribute

An item’s description

Why is OOP useful?

Makes translating the real world easier

- Humans naturally think about the world in terms of objects and responsibilities
  - You don’t try to make a call from your hand - because your hand does not have that function.



Most things are built layer-by-layer

- Engines have engine blocks, pistons, pipes, pumps, belts, gears, screws. Each of these things added one on top of the other creates the whole.

Worked Example

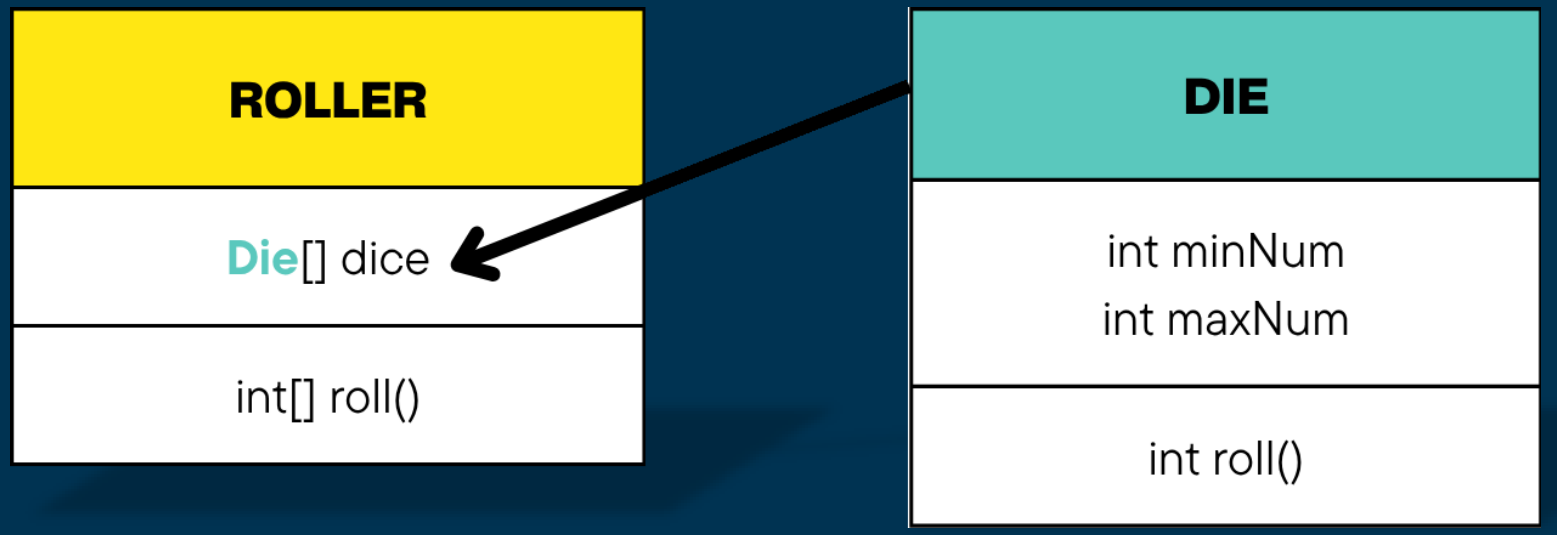
Dice Roller Game

A client wants us to create a die roller game. They want to roll a given number of dice for a given number of players, record the rolls & want the ability to see who rolled the highest & lowest.



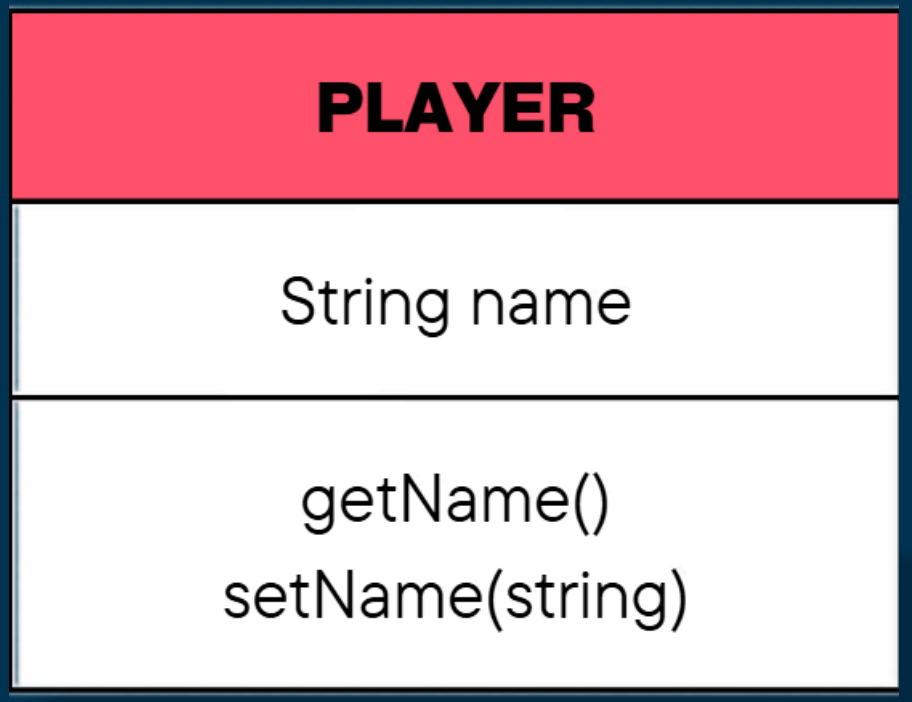
Dice Roller Game

"They want to roll a given number of dice"



Dice Roller Game

“for a given number of players”



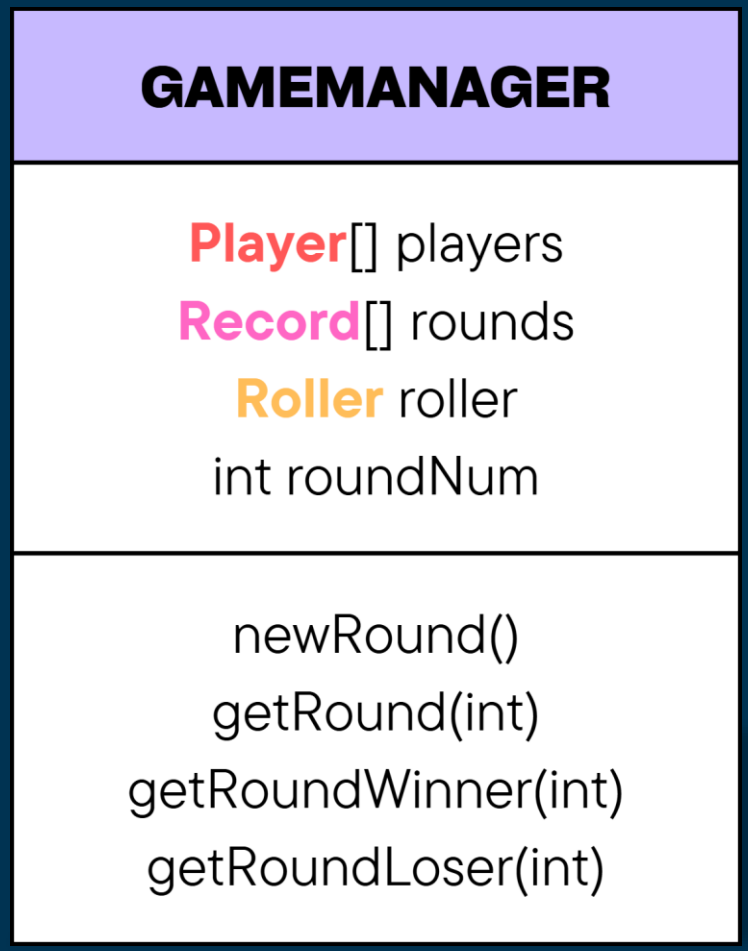
Dice Roller Game

“record the rolls”



Dice Roller Game

“who rolled the highest & lowest”

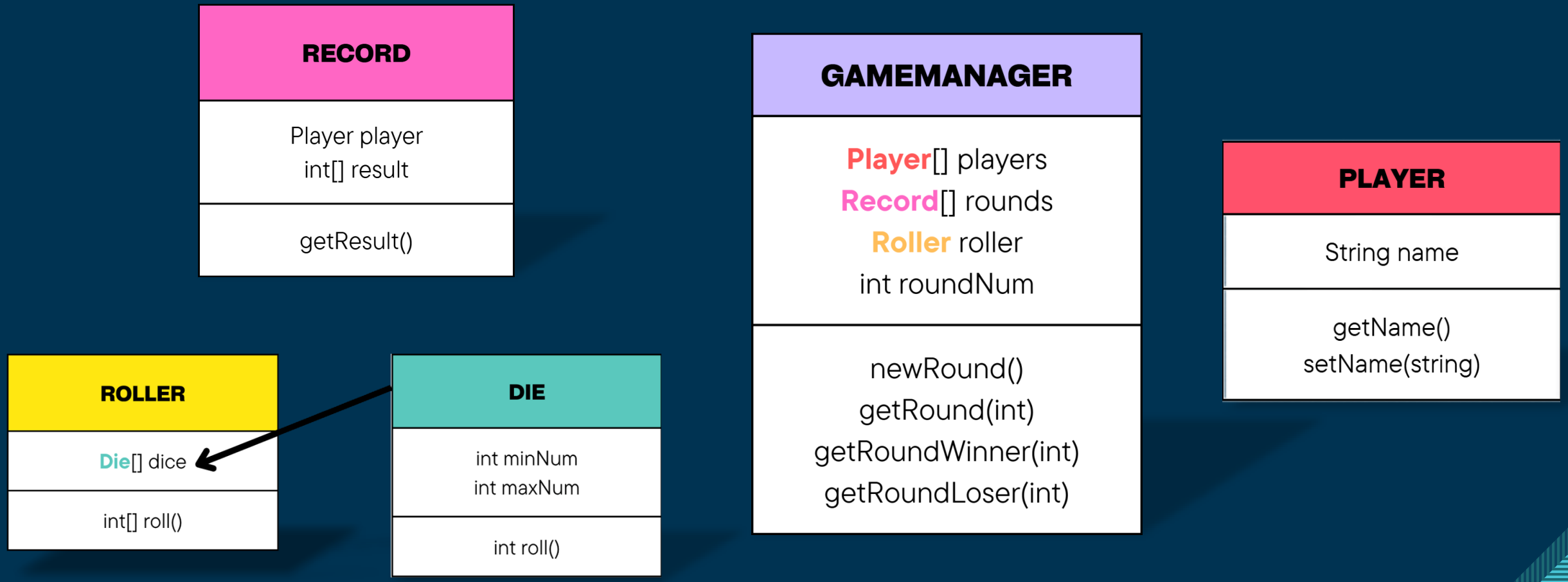


Dice Roller Game

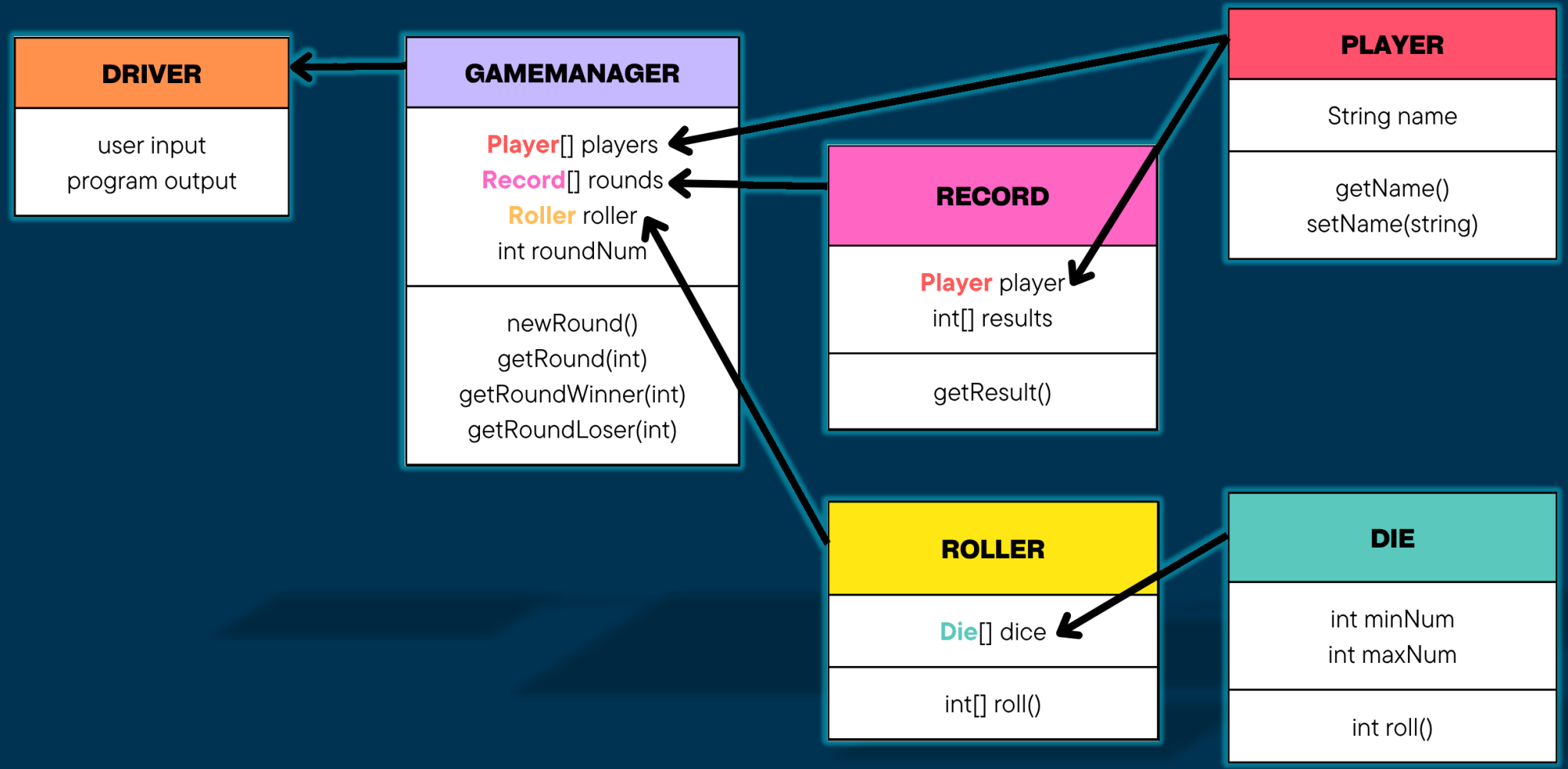
A client wants us to create a die roller game. They want to roll a given number of dice for a given number of players, record the rolls & want the ability to see who rolled the highest & lowest.



Dice Roller Game



Putting Everything Together



Class Exercise

Groups of 3-4

Custom Car Dealership

A custom car dealership allows customers to build a car by making an order: choosing an engine, body type, colour and wheels. The dealership wants you to build this system.



Custom Car Dealership

A custom car dealership allows customers to build a car by making an order: choosing an engine, body type, colour and wheels. The dealership wants you to build this system.



Thank You

From Idea to Implementation

Christopher Bolger
3rd Year Computer Science
Game Development